images are run on the Apple Silicon computers (ARM64) through a compatibility layer based on Rosetta, which impacts application startup time.

Go ahead and configure the liveness and readiness probes in the Deployment manifests for all the applications composing the Polar Bookshop system. As a reference, you can look at the source code repository accompanying this book (Chapter13/13-end).

On top of event logs, health information improves the information we can infer about the application's internal state, but it's not enough to achieve complete visibility. The following section will introduce the concept of metrics and how we can configure them in Spring Boot.

13.3 Metrics and monitoring with Spring Boot Actuator, Prometheus, and Grafana

To properly monitor, manage, and troubleshoot an application running in production, we need to be able to answer questions like "how much CPU and RAM is the application consuming?", "how many threads are used over time?", and "what's the rate of failing requests?" Event logs and health probes can't help us answer those questions. We need something more. We need more data.

Metrics are numeric data about the application, measured and aggregated in regular time intervals. We use metrics to track the occurrence of an event (such as an HTTP request being received), count items (such as the number of allocated JVM threads), measure the time taken to perform a task (such as the latency of a database query), or get the current value of a resource (such as current CPU and RAM consumption). This is all valuable information for